

Sparse Blind Deconvolution via Thresholded Wirtinger Flow

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Background and Problem Formulation

• **Observation and sparse priors.** We observe $\mathbf{f} \in \mathbb{C}^M$, the length- M circular convolution of an unknown s_h -sparse kernel $\mathbf{h} \in \mathbb{C}^M$ and a latent signal $\mathbf{g} = \mathbf{C}\mathbf{x}^*$. Here $\mathbf{C} \in \mathbb{C}^{M \times N}$ is a known dictionary, $\mathbf{x} \in \mathbb{C}^N$ is an unknown s_x -sparse coefficient vector, and $\otimes$ denotes length- M circular convolution:

$$\mathbf{f} = \mathbf{h} \otimes \mathbf{g} = \mathbf{h} \otimes \mathbf{C}\mathbf{x}^*, \quad \|\mathbf{h}\|_0 \leq s_h, \quad \|\mathbf{x}\|_0 \leq s_x.$$

• **Fourier-domain bilinear model.** Let $\mathbf{B}^H$ be the unitary DFT matrix, $\mathbf{y} = \mathbf{B}^H \mathbf{f}$, and $\mathbf{A} := \sqrt{M} \mathbf{C}^T \mathbf{B}^*$. If $\mathbf{a}_j$ and $\mathbf{b}_j$ denote the j th columns of $\mathbf{A}$ and $\mathbf{B}$, respectively, then

$$y_j = \mathbf{b}_j^H \mathbf{h} \mathbf{x}^H \mathbf{a}_j + e_j, \quad \mathbf{e} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_M).$$

Thus $\mathbf{y}$ contains the frequency-domain measurements; the noiseless model is recovered by setting $\mathbf{e} = \mathbf{0}$.

• **Sparse least-squares estimator.**

$$\begin{aligned} & \underset{\mathbf{h}, \mathbf{x}}{\text{minimize}} \quad \sum_{j=1}^M |\mathbf{b}_j^H \mathbf{h} \mathbf{x}^H \mathbf{a}_j - y_j|^2 \\ & \text{subject to} \quad \|\mathbf{h}\|_0 \leq s_h, \quad \|\mathbf{x}\|_0 \leq s_x. \end{aligned}$$

Identifiability. Under the stated sensing and sparsity assumptions, this problem is identifiable up to the intrinsic scaling ambiguity

$$(\mathbf{h}, \mathbf{x}) \sim (\mathbf{h}/\alpha^*, \alpha \mathbf{x}), \quad \alpha \in \mathbb{C} \setminus \{0\}.$$

Every pair in this equivalence class produces exactly the same measurements $\mathbf{y}$. Consequently, recovery and convergence are evaluated modulo this unavoidable scale ambiguity, as reflected by the distance used below.

Proposed Two-Stage Method

Stage 1: Sparse spectral initialization. Construct the adjoint matrix

$$\mathbf{Z} = \sum_{j=1}^M y_j \mathbf{b}_j \mathbf{a}_j^H, \quad \mathbb{E}[\mathbf{Z}] = \mathbf{h}^* \mathbf{x}^{*H}.$$

Compute $c_l = \max_{i \in [N]} |Z_{l,i}|$ and $r_i = \max_{l \in [M]} |Z_{l,i}|$; the largest s_h values of c_l and largest s_x values of r_i define $\bar{I}_h$ and $\bar{I}_x$. On the restricted matrix, take the singular value decomposition

$$\mathbf{Z}_{\bar{I}_h, \bar{I}_x} = \mathbf{U} \Sigma \mathbf{V}^H, \quad (\sigma_1, \mathbf{p}, \mathbf{q}) = (\Sigma_{11}, \mathbf{U}_{:,1}, \mathbf{V}_{:,1}),$$

and zero-pad the leading singular vectors:

$$\mathbf{h}_{\bar{I}_h}^0 = \sqrt{\sigma_1} \mathbf{p}, \quad \mathbf{x}_{\bar{I}_x}^0 = \sqrt{\sigma_1} \mathbf{q}.$$

Stage 2: Thresholded Wirtinger Flow (ThWF). Normalize each gradient by the energy of the paired factor, then project back to the sparse model:

$$\mathbf{h}^{k+1} = \mathcal{T}_{s_h} \left(\mathbf{h}^k - \frac{\eta}{\|\mathbf{x}^k\|^2} \nabla_{\mathbf{h}} f(\mathbf{x}^k, \mathbf{h}^k) \right), \quad \mathbf{x}^{k+1} = \mathcal{T}_{s_x} \left(\mathbf{x}^k - \frac{\eta}{\|\mathbf{h}^k\|^2} \nabla_{\mathbf{x}} f(\mathbf{x}^k, \mathbf{h}^k) \right).$$

Here $\mathcal{T}_s(\mathbf{v})$ retains the s largest-magnitude entries of $\mathbf{v}$ and sets all remaining entries to zero. Scale normalization balances the bilinear factors, while hard thresholding keeps every iterate sparse.

Main Theoretical Guarantees

For $\mathbf{z}^k = [\mathbf{h}^{kH}, \mathbf{x}^{kH}]^H$, define the scaling-invariant distance

$$\text{dist}(\mathbf{z}^k, \mathbf{z}^*) = \min_{\alpha \in \mathbb{C}} \sqrt{\|\mathbf{h}^k/\alpha^* - \mathbf{h}^*\|^2 + \|\alpha \mathbf{x}^k - \mathbf{x}^*\|^2}.$$

Assumptions.

- **(A1) Gaussian sensing:** $\mathbf{a}_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$.
- **(A2) Spectral flatness:** $\max_j |\mathbf{b}_j^H \mathbf{h}^*| \leq \mu \|\mathbf{h}^*\|/\sqrt{M}$, where $1 \leq \mu \leq \sqrt{M}$.
- **(A3) Minimum amplitudes:** $\min_{l \in \bar{I}_h} |\mathbf{h}_l^*| \geq C_h \|\mathbf{h}^*\|/\sqrt{s_h}$ and $\min_{i \in \bar{I}_x} |x_i^*| \geq C_x \|\mathbf{x}^*\|/\sqrt{s_x}$. In noise, their product is at least $C' \sigma \sqrt{\log(MN)} \|\mathbf{h}^*\| \|\mathbf{x}^*\|$.

Theorem 1 (Sparse spectral initialization). Under (A1)–(A3), if $M \geq C_0 s_h s_x \log M$, Algorithm 1 exactly recovers the supports of $\mathbf{h}^*$ and $\mathbf{x}^*$ with high probability. Moreover, under (A1)–(A2), if $M \geq C'_0 (s_h^2 + \mu^4 s_x^2) \log M$, then, with high probability, the initial point $\mathbf{z}^0 = [\mathbf{h}^{0H}, \mathbf{x}^{0H}]^H$ satisfies

$$\text{dist}(\mathbf{z}^0, \mathbf{z}^*) \leq \lambda \|\mathbf{z}^*\|, \quad 0 < \lambda < 1.$$

Theorem 2 (Local linear convergence). Suppose (A1)–(A3) hold, $\mathbf{z}^0$ is the sparse spectral initializer, and $\{\mathbf{z}^k\}_{k \geq 0}$ is generated by ThWF from noiseless measurements. Choose $\eta \leq 1$. If $M \geq C_1 \mu^2 (s_h + s_x) \log^9 M$, then, with high probability,

$$\text{dist}(\mathbf{z}^{k+1}, \mathbf{z}^*) \leq (1 - \eta/16) \text{dist}(\mathbf{z}^k, \mathbf{z}^*), \quad \forall k \geq 0.$$

Theorem 3 (Noisy local linear convergence). Suppose (A1)–(A3) hold, $\mathbf{z}^0$ is the sparse spectral initializer, and $\{\mathbf{z}^k\}_{k \geq 0}$ is generated from noisy measurements. Choose $\eta \leq 1$ and suppose $\sigma \leq C_\sigma \|\mathbf{x}^*\| \|\mathbf{h}^*\|/\sqrt{M \log M}$, and $M \geq C_2 \mu^2 (s_h + s_x) \log^9 M$. With high probability,

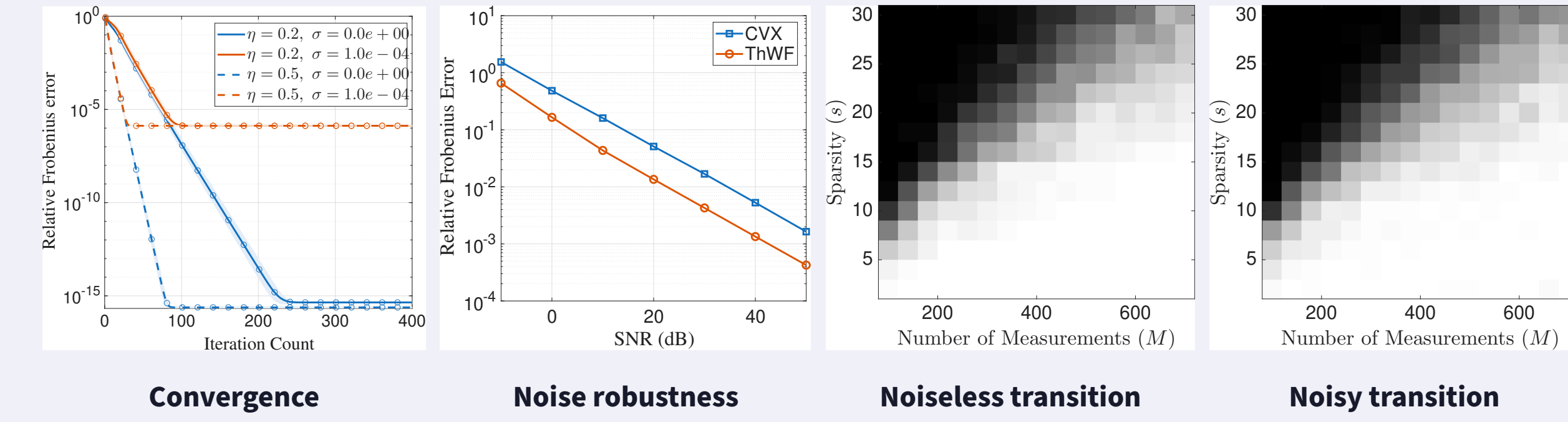
$$\text{dist}^2(\mathbf{z}^{k+1}, \mathbf{z}^*) \leq (1 - \eta/16) \text{dist}^2(\mathbf{z}^k, \mathbf{z}^*) + \mathcal{O}(\eta M \sigma^2), \quad \forall k \geq 0.$$

Interpretation. Sparse spectral initialization enters a local contraction basin. In the noiseless case ThWF converges geometrically to the ground truth up to scaling; with noise, it contracts to a limiting squared error of order $\mathcal{O}(M \sigma^2)$.

Limitation. Although exact support recovery needs only $\tilde{\mathcal{O}}(s_h s_x)$ samples, our initializer still requires $\tilde{\mathcal{O}}(s_h^2 + \mu^4 s_x^2)$ samples to guarantee an accurate warm start. Conditional on the correct supports, the unknown rank-one factors have only $s_h + s_x$ degrees of freedom, so an additive $\tilde{\mathcal{O}}(s_h + s_x)$ SVD-estimation bound is statistically plausible and consistent with the additive minimax rate in sparse CCA [1]. Moreover, since μ chiefly affects recovery of $\mathbf{x}$, the asymmetric scaling $\tilde{\mathcal{O}}(s_h + \mu^2 s_x)$ may be sharper than $\tilde{\mathcal{O}}(\mu^2 (s_h + s_x))$.

Experiments

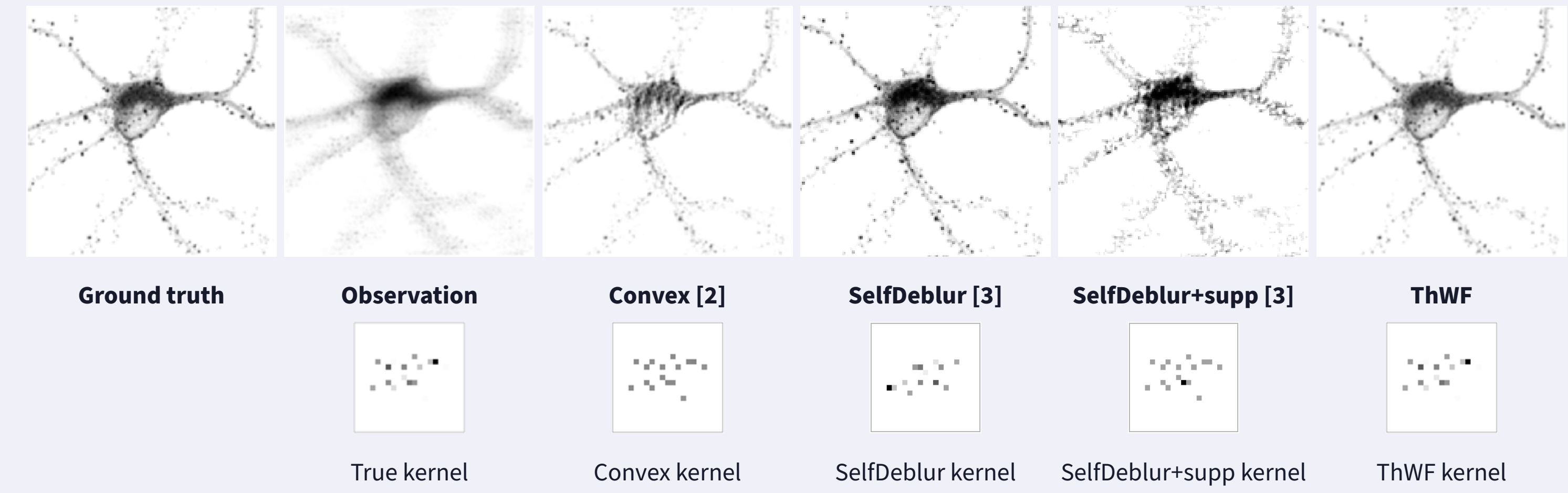
• **Synthetic experiments.** We run 100 Monte Carlo trials per configuration. Recovery terminates when the relative lifted error falls below 10^{-8} or after 500 iterations.



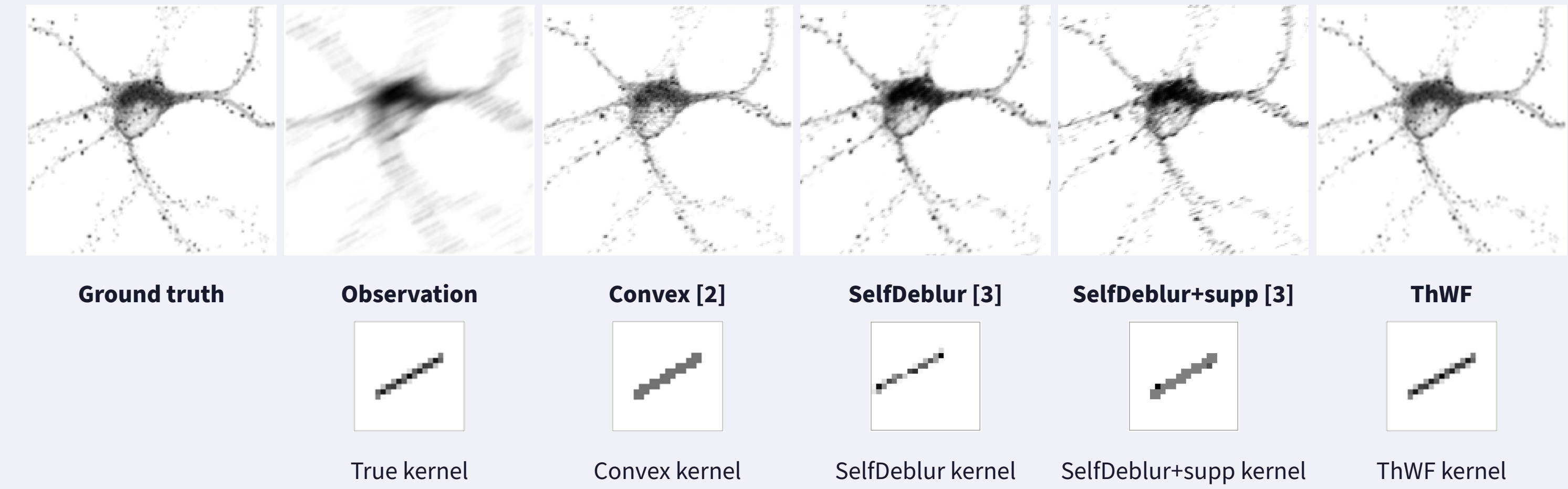
The experiments match the theory: ThWF contracts linearly, its final error scales with the noise level, and successful recovery exhibits a sharp transition as the measurement count grows relative to sparsity.

• **Real image deblurring experiments.** We compare ThWF with convex lifting [2] and SelfDeblur [3] on a 160×160 neuron image under Bernoulli-Gaussian (BG), motion, and Gaussian blur. SelfDeblur w/supp [3] is given the true kernel support.

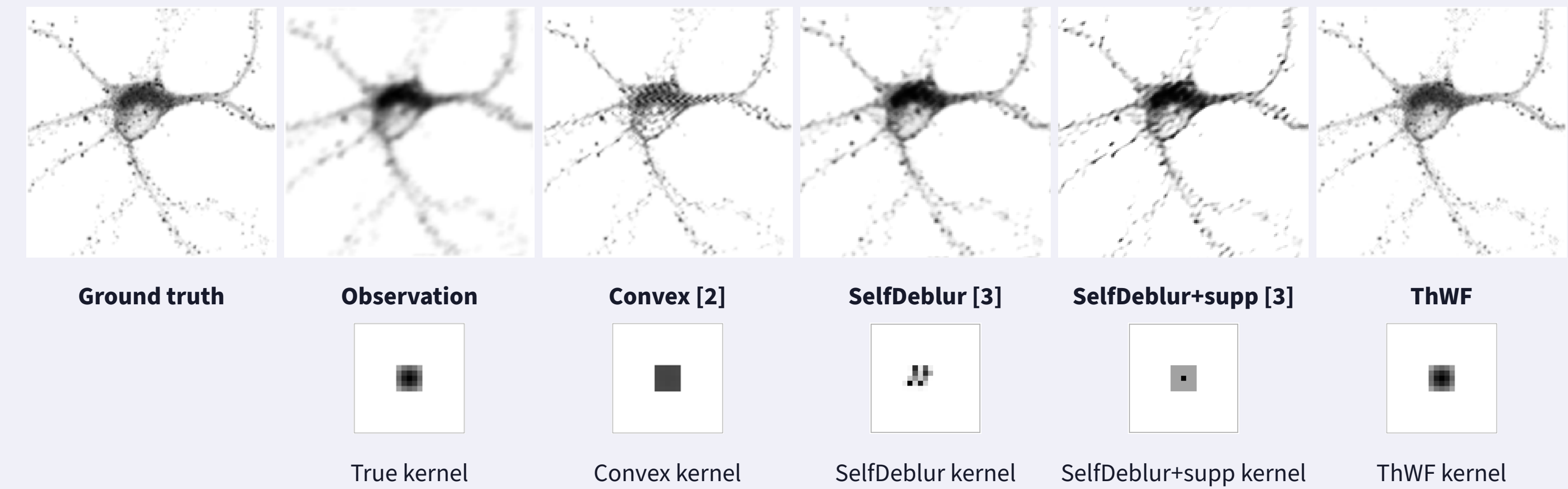
BG kernel case



Motion kernel case



Gaussian kernel case



PSNR (dB)	BG	Motion	Gaussian
Convex lifting [2]	25.82	31.48	28.69
SelfDeblur [3]	22.88	21.89	26.25
SelfDeblur w/supp [3]	20.99	27.11	27.14
ThWF	111.10	83.40	40.18

ThWF achieves the highest PSNR for all three kernels and accurately recovers both the latent image and blur kernel.

References

- [1] C. Gao, Z. Ma, and H. H. Zhou, "Sparse CCA: Adaptive Estimation and Computational Barriers," *Ann. Statist.*, vol. 45, no. 5, pp. 2074–2101, 2017.
- [2] A. Ahmed, B. Recht, and J. Romberg, "Blind deconvolution using convex programming," *IEEE Trans. Inf. Theory*, vol. 60, no. 3, pp. 1711–1732, 2014.
- [3] D. Ren, K. Zhang, Q. Wang, Q. Hu, and W. Zuo, "Neural Blind Deconvolution Using Deep Priors," *Proc. CVPR*, pp. 3341–3350, 2020.